Kuan Heng (Jordan) Lin

jordan@cs.columbia.edu | kuanhenglin | kuanhenglin.github.io

Research Interests

Computer vision, multimodal generative models, 3D/4D reconstruction, visual reasoning, world models & representation learning

Education

Columbia University in the City of New York *August 2025–Present*
Ph.D. in Computer Science
Advisors: Aleksander Hołyński, Micah Goldblum

University of California, Los Angeles *September 2021–June 2025*
B.S. in Computer Science, minor in Mathematics, *Summa Cum Laude* (GPA: 3.980)
Advisor: Bolei Zhou

Publications

[1] Kuan Heng Lin, Zhizheng Liu, Pablo Salamanca, Yash Kant, Ryan Burgert, Yuancheng Xu, Koichi Namekata, Yiwei Zhao, Bolei Zhou, Micah Goldblum, Paul Debevec, Ning Yu. Vista4D: Video Reshooting with 4D Point Clouds. In: *Computer Vision and Pattern Recognition (CVPR)*, **Highlight**. 2026.

[2] Koichi Namekata, Yash Kant, Zhizheng Liu, Ryan Burgert, Yuancheng Xu, Kuan Heng Lin, Emmett Steven, Julien Philip, Li Ma, Andrea Vedaldi, Paul Debevec, Ning Yu. Go-with-the-Track: Video Compositing and Motion Control with Point Tracking. In: *ACM Transactions on Graphics (SIGGRAPH)*. 2026.

[3] Kuan Heng Lin*, Sicheng Mo*, Ben Klingher, Fangzhou Mu, Bolei Zhou. Ctrl-X: Controlling Structure and Appearance for Text-To-Image Generation Without Guidance. In: *Neural Information Processing Systems (NeurIPS)*. 2024.

[4] Sicheng Mo*, Fangzhou Mu*, Kuan Heng Lin, Yanli Liu, Bochen Guan, Yin Li, and Bolei Zhou. FreeControl: Training-Free Spatial Control of Any Text-to-Image Diffusion Model with Any Condition. In: *Computer Vision and Pattern Recognition (CVPR)*. 2024.

Research Experiences

Graduate Research Assistant | Department of Computer Science, Columbia University *September 2025–Present*

- Working on visual reasoning (in generative models), multimodal 3D/4D reconstruction and VLMs, and (visual) generative models.

Research Intern | Netflix Eyeline Labs *March 2025–August 2025*

- Work on video novel view synthesis and 4D scene editing (Vista4D, CVPR 2026 Highlight), along with video motion control and reference-based editing (Go-with-the-Track, SIGGRAPH 2026).
- Develop a framework to reshoot a video by moving the camera and manipulating the scene in a 4D point cloud, which is robust to real-world 4D reconstruction artifacts to bridge the train-inference gap and enable cool applications (scene editing and expansion).

Undergraduate Researcher | Zhou Lab at UCLA *March 2023–March 2025*

- Design a lightweight, universal training-free and guidance/optimization-free structure and appearance control for text-to-image generation with modality-agnostic structure preservation and semantically-aware appearance transfer (Ctrl-X, NeurIPS 2024).
- Propose training-free controllable and image-to-image generation for any text-to-image model via principal component extraction of seed images and diffusion guidance optimization with arbitrary condition images (FreeControl, CVPR 2024).

Research Engineering Intern | Snap Inc. *June 2024–September 2024*

- Work in the Creative Vision team to optimize training and inference of large-scale video generation with `PyTorch` profiling and parallelism.
- Implement pipeline-wide intra-batch variable sequence length training for arbitrary datasets, modalities, and conditioning for flexible, efficient multimodal training while maintaining parallelisms, performance, and memory usage of non-variable-length training.
- Build Snap's first distributed 2D parallel (Data + Model Parallel) training framework for scaling video Diffusion Transformers with `torch.distributed`, designing custom Tensor Parallel & Sequence Parallel strategies and wrappers for video DiTs and convolutional autoencoders, significantly scaling up model size and number of tokens while minimizing VRAM usage and GPU communication overhead.

Undergraduate Researcher | The Ozcan Research Group (HHMI Program) *October 2022–June 2023*

- Design Fourier residual and attention blocks with depthwise separable convolutions and linear discrete cosine transform for Diffusion Models that maintain baseline performance while having significantly fewer parameters and accelerated training.
- Adapt ResNet and ResNeXt architectures with Fourier residual networks to learn image features in both the discrete and frequency domain, achieving improved classification accuracy compared to baselines with fewer parameters.

Other Work Experiences

Program Development Team | UCLA CS, UCLA CAE, Learning Assistant Program *June 2023–June 2025*

- Work with professors & TAs to expand the LA Program to support high-demand upper-division courses (e.g., CS 111, 118, 131), helping them implement evidence-based pedagogy in CS classrooms and empowering students through collaborative and inclusive teaching.
- Oversee all CS course program operation and advertise the LA program to increase applicants for CS courses by 300%.
- Optimize & automate LA application and review with Airtable `JavaScript` and Gmail scripting for 900+ applicants and supporting 14000+ students in UCLA STEM courses, streamlining applicant review and communication.

Head Learning Assistant | UCLA CS, UCLA CAE, Learning Assistant Program *March 2023–June 2023*

- Lead weekly discussions and workshops for CS 33: Computer Organization to review material and lead worksheets for 400+ students. Reviewed very positively, notably my willingness to help, clarity of explanations, and passion for teaching.
- Organize meetings, plan pedagogy activities and workshops, and host content meetings to facilitate and mentor CS 33 LAs.

Projects, Volunteering, & Service

Conference Reviewer | CVPR 2025–26, ECCV 2024/26, NeurIPS 2025–26, SIGGRAPH Asia 2026, BMVC 2026 *May 2024–Present*

Advisor ← Co-President ← Workshops Officer | ACM Student Chapter at UCLA, AI Committee *May 2022–June 2025*

- Founded weekly reading groups & seminars discussing recent ML advances such as generative vision, reinforcement learning, and LLMs with student & industry speakers, culminating in projects and events for general members such as the adversarial AI competition.
- Spearhead general member programs (e.g., special topics discussions) and bold initiatives (e.g., AI hackathons, research team, shared compute, inter-comittee collaborations) which drastically improved member retention, officer burnout, and club exposure.
- Revamp and teach workshops on deep learning topics such as gradient descent, backpropagation, neural networks, CNNs, RNNs, Transformers, generative vision, `Python` package management, and hands-on notebooks, improving retention by 100%.

Assistant Managing Editor of Review | UCLA Undergraduate Science Journal *January 2022–June 2025*

- Draft and polish letters to authors which succinctly culminate and summarize all reviewer reviews, provide constructive and detailed feedback for authors, and prepare comprehensive reports for the editorial board for further communication.
- Lead small teams of reviewers to set rigorous benchmarks and helpful guidelines for reviewing papers.

Cinematographer, Editor, Composer | UCLA Film & Photography Society Production *September 2021–March 2023*

- *Manic Pixie Vending Machine* (cinematographer, colorist, assistant editor), *Fear No Evil* (cinematographer, colorist, assistant editor), *On Her Time* (cinematographer, colorist, assistant editor, composer), *Doldrums* (behind-the-scenes cinematographer).
- Lead large camera & lighting teams to set up lighting, camera equipment, and sets to create stunning visuals from little budget.
- Used professional cameras, gimbals, and sliders, colorgraded and edited Adobe Premiere projects with sound mixing and composition.

Awards & Honors

Phi Beta Kappa Honor Society *June 2025*
Undergraduate Research Scholars Program ($6000) *September 2024–June 2025*
CRA Outstanding Undergraduate Researcher Award Finalist (UCLA CS Newsroom, CRA Q&A Highlight) *January 2025*
NeurIPS Scholar Award *December 2024*
URC-Sciences Summer Program ($6000) *June 2023–September 2023*
UCLA Dean's Honors List *Fall 2021–Spring 2025*
Upsilon Pi Epsilon Honor Society *December 2022*

Skills

Programming: `Python` (`PyTorch`, `TensorFlow`, `JAX`), `CUDA`, `C++`, `C`, `Assembly`, `Verilog`, `R`, `Haskell`, `React`, `Flask`, `JavaScript`, `HTML/CSS`
Miscellaneous: Pedagogy, science communication, filmmaking, photography, video editing, content creation, music composition
Languages: English (native), Chinese (native)